Experiment No.1

Title: Perform following String operations: a.substring b. palindrome c. compare d. copy e. Reverse

Name of the student:-Gauravi Rajpal Shinde
Class:-SY1 Batch:-B1 PRN:B25ET1046 Div:-1

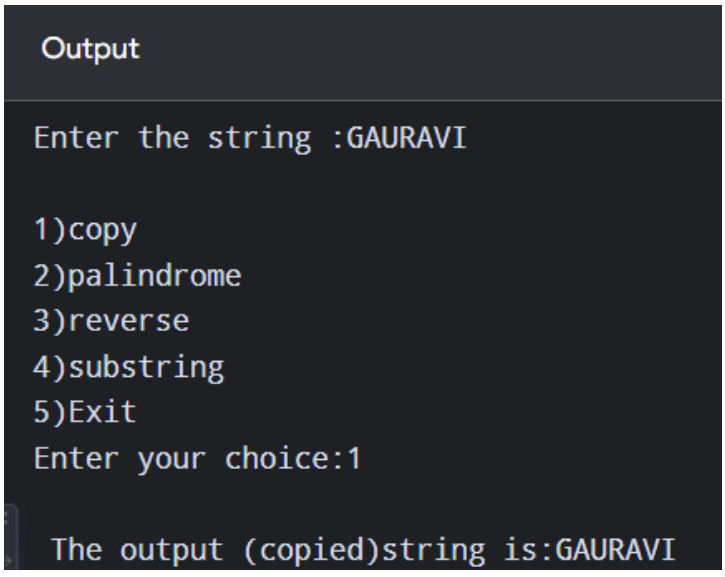
INPUT Program:

```
#include<stdio.h>
#include<string.h>
void reverse(char input[]);
int palindrome(char input[]);
void copy(char input[],char output[]);
void substring(char input[]);
int stringlength(char input[]);
int main()
{
char str[10],output[10], temp;
int ch,c,i,n;
printf("\nEnter the string :");
scanf("%s",str);
do
{
printf("\n1)copy\n2)palindrome\n3)reverse\n4)substring\n5)Exit");
printf("\nEnter your choice:");
scanf("%d",&ch);
switch(ch)
{
case 1: copy(str,output);
break;
case 2:
c=palindrome(str);
if(c==1)
printf("The given string is a Palindrome");
else
printf("The given string is not a palindrome");
break;
case 3: reverse(str);
break;
case 4: substring(str);
}
}while(ch!=5);
}
```

```
int palindrome(char name[])
{
int i=0,j=0;
while(name[j]!='\0')
j++;
j--;
while(i<j)
{
if(name[i]!=name[j])
return(0);
i++;
j--;
}
return(1);
}
void copy(char input[],char output[])
{
int i;
for(i=0;input[i]!='\0';i++)
output[i]=input[i];
output[i]='\0';//append the termination character
printf("\n The output (copied)string is:");
printf("%s",output);
}
void substring(char input[])
{
int i,n,position;
char output[10];
printf("Enter the position of substring:");
scanf("%d",&position);
n=stringlength(input);
if(position<n)
{
for(i=0;i<n-position;i++)
output[i]=input[position+i];
output[i]='\0';
printf("\n input string:%s \noutput substring:=%s", input,output);
}
else
printf("\n position entered is out of range");
}
int stringlength(char input[])
{
```

```
int i=0;
while(input[i]!='\0')
i++;
return i;
}
void reverse(char input[])
{int n,i;
char output[10];
n=stringlength(input);
for(i=0;i<n;i++)
output[n-1-i]=input[i];
output[i]='\0';
printf("\n input string:%s \noutput (reversed)String:=%s", input,output);
}
```

OUTPUT:



Output

```
 The output (copied)string is:GAURAVI
1)copy
2)palindrome
3)reverse
4)substring
5)Exit
Enter your choice:2
The given string is not a palindrome
1)copy
2)palindrome
3)reverse
4)substring
5)Exit
Enter your choice:3

 input string:GAURAVI
output (reversed)String:=IVARUAG
```

```
1)copy
2)palindrome
3)reverse
4)substring
5)Exit
Enter your choice:4
Enter the position of substring:4

 input string:GAURAVI
output substring:=AVI
1)copy
2)palindrome
3)reverse
4)substring
5)Exit
Enter your choice:5


=== Code Execution Successful ===
```